

loud, some weak, some held for long intervals, some held for short intervals, and resultant speech that is often unintelligible.

- Failure of progression occurs in talking because the formation of words depends on rapid and orderly succession of individual muscle movements in the larynx, mouth, and respiratory system.

viii. Unsteady of gait.

- The gait is unsteady and broad-based due to dysmetria and kinetic tremors of the lower limbs.

9. Functions of vestibular apparatus

The maintenance of the erect posture is mediated through two antagonistic forces:

- The force of gravity.
- The tonic contraction of antigravity muscles.
- The center of gravity changes continuously according to body posture.
- The postural reflexes increase the tone of antigravity muscles to maintain the erect posture each time the body position changes.
- The body has no receptors for constant speed motion. We do not feel the constant speed movement of earth at the high speed of 30 Km/second.
- The body can detect:
 - A. Movement with changing rate (acceleration).
 - B. Movement of changing direction which may be: linear or angular.

THE VESTIBULAR APPARATUS (figure 38)

- The VA of the inner ear is present in the petrous part of the temporal bone.
- When stimulated, it sends impulses to inform the brain

about: Orientation of the head in space.

Linear or angular accelerations.

- It is formed of:
 1. The bony labyrinth: the cochlea, vestibule and 3 SCCs.
 2. The membranous labyrinth: they are membranous tubes inside the bony labyrinth filled with endolymph and surrounded with perilymph. It is divided into two parts:

Auditory labyrinth: cochlea.

Non-auditory labyrinth: 3 SCCs, Utricle and Saccule.

The Semicircular Canals (SCCs):

- They are arranged in 3 planes (one horizontal and two vertical) in space each is perpendicular to the other two.
- The horizontal SCC:
 - It is not exactly horizontal but inclined backward 30° in the normal head position. Ventroflexion of the head 30° makes it absolutely horizontal.
- The right and left horizontal SCCs are in the same plane.
- The vertical SCCs (anterior and posterior):
 - They are perpendicular to each other.

- The anterior vertical SSC of one side is in the same plane as the posterior vertical SCC of the opposite side and each makes an angle of 45° with the sagittal and frontal planes.
- Each SCC has a dilated end called ampulla containing the receptor for angular movement (crista ampullaris) and a non ampullary end.
- The 3 SCCs open into the corresponding Utricle by only 5 openings because the non ampullary ends of the vertical SCCs fuse and open by only one opening.

The structure of the SCCs:

A cross section through the SCC shows:

- The membranous labyrinth is formed of 3 layers: outer vascular, middle connective tissue (substantia propria) and inner folded flat endothelium.

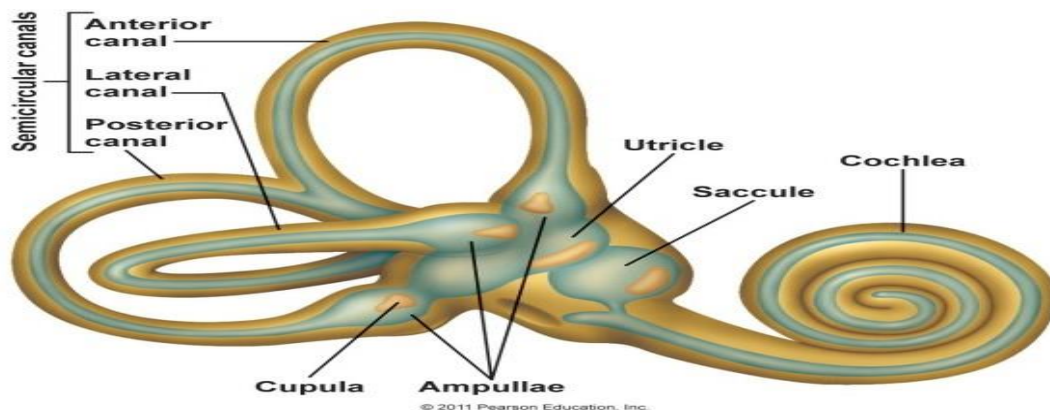


Figure 38 : The vestibular Apparatus

The crista ampullaris (figure 39):

- The substantia propria forms a ridge covered by columnar epithelial cells; some are supporting and other are receptor cells.

- Each receptor columnar cell has short hairs (stereocilia, 40-70) and one long kinocilium at the peripheral luminal margin.
- The cilia project in the lumen surrounded by gelatinous material forming the cupola which touches the opposite wall of the canal dividing the endolymph into 2 separate partitions.
- The cupola can be deflected by movement of the endolymph within the canal.

N.B.:

* Displacement of stereocilia towards the kinocilium stimulates and away from it inhibits the receptor.

* In the horizontal SCC, the kinocilium lies towards the Utricle, but the reverse in the vertical SCC.

* Impulses from the crista ampullaris are conducted along the vestibular nerve.

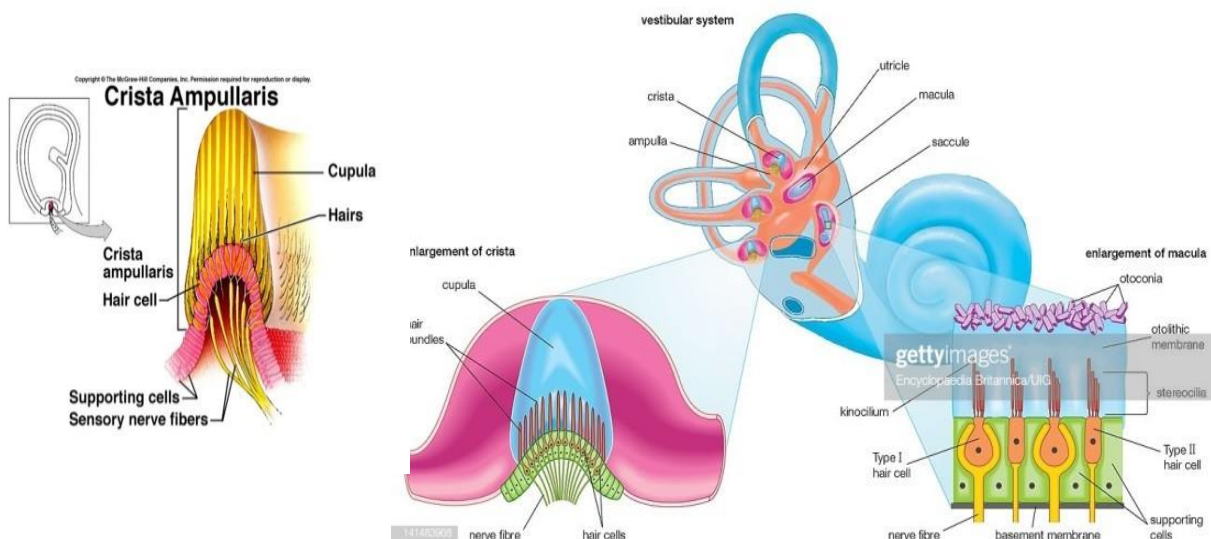


Figure 39 crista ampullaris

Mechanism of stimulation of the SCC

Stimulus: angular rotation by a revolving chair.

The stimulated canal: horizontal SCC:

At rest: there is no discharge from the crista of the horizontal SCCs of either side and no sense of rotation.

At the start of movement:

- The bony labyrinth moves from right to left as the direction of rotation.
- The endolymph lags behind the movement of the bony labyrinth because of its inertia (high viscosity) → behaves as if it moves from left to right →
- Displaces the cupola of the right crista towards the Utricle → stimulation.
- Displaces the cupola of the left crista away from the Utricle → inhibition.
- Unequal discharge from either sides → sense of rotation to the right.

During continuous movement:

- The inertia of endolymph is overcome → moves the same rate as and direction as the bony labyrinth → cupola on either sides returns to a central position → equal slight or no discharge from either sides → no sense of rotation.

When movement is stopped:

- The bony labyrinth stops movement; while the endolymph continues movement from right to left by its inertia →
- Displaces the cupola of the right crista away from the Utricle → inhibition.
- Displaces the cupola of the left crista towards the Utricle → stimulation.
- Unequal discharge from either sides → sense of counter rotation.
- The cupola on either sides returns to a central position → equal slight

or no discharge from either sides → no sense of rotation.

Methods of stimulation of SCCs

I. Rotation Method:

By putting the person on a revolving chair (10 times/seconds and the stimulated SCC in the horizontal plane (table 15):

SCC stimulated	Head position
Horizontal	Ventroflexed 30°
Vertical	Dorsiflexed 60°

II. Thermal Method:

By douching the external ear with either warm (43°) or cold (20°) water. the SCC in the vertical plane is the canal stimulated (table 16):

SCC stimulated	Head position
Horizontal	Dorsiflexed 60°
Vertical	Normal position

III. Galvanic Stimulation:

- By placing the stimulating electrode over the mastoid process.
- This stimulates the vestibular nerve directly and test its integrity, but not the SCCs.

Neuronal connection of the SCC

Impulses are relayed from the crista ampullaris → vestibular nerve → vestibular ganglion → enters the brain stem at the level of pons to connect to:

1. Vestibulocerebellar fibers:

Direct: fibers enter through the ICP → flocculonodular (archicerebellum)

Indirect: vestibular fibers end on vestibular nuclei in medulla → ICP → archicerebellum.

Fate: Dentate nucleus → SCP → thalamus of opposite side (either directly or after relay in RN) → to CC →

- Superior temporal gyrus (equilibrium center) → sensation of rotation.
- Motor cortex → control of motor functions and muscle tone according to labyrinthine sensory impulses.

2. Vestibulothalamic fibers:

Vestibular nuclei → reticular nuclei → thalamus of opposite side → superior temporal gyrus.

Function:

- A. Another pathway for the sense of rotation.
- B. Conscious perception of head position.

3. Vestibuloreticular fibers:

Vestibular nuclei → reticular nuclei of the brain stem.

Function:

- A. Production of autonomic effects of rotation e.g. vomiting.
- B. Control of muscle tone through the reticulospinal tract.

Effect of stimulation of SCCs

1) Nystagmus

- Definition: Oscillatory movement of the eye ball consisting of slow and rapid components to fix objects in the visual field.

- Cause: the most common is post rotatory.
- Types: Horizontal. -Vertical. -Rotatory.
- Direction of nystagmus:
 1. The slow component: is opposite to the direction of rotation.
 2. The rapid component: is in the direction of rotation and nystagmus is named according to it e.g. left horizontal nystagmus means horizontal nystagmus with the rapid component to the left.

Pathway:

- The slow component: Crista → vestibular nerve → vestibular nuclei → MLB of both sides → extrocular muscles.
- The rapid component: RF → MLB → of both sides → extraocular

muscles Causes of nystagmus

1. *Physiological nystagmus:*

It occurs in normal individuals and include:

- A. Post-rotatory due to angular rotation.
- B. Ocular or Optokinetic:

Cause: looking at continuously moving field e.g. from a window of a care or a train.

Function: to fix successive objects in the visual field.

Pathway: Retina → visual pathway → occipital cortex → internal capsule → lateral part of cerebral peduncles → MLB → extraocular muscles

- C. Spontaneous: caused by looking at an object in the extreme lateral part of the visual field.

2. *Pathological nystagmus:*

i. Labyrinthine lesions:

This leads to unequal discharge from both labyrinths → sense of rotation towards the more stimulated side.

Complete damage of both labyrinths → no discharge from both sides → no sense of rotation.

ii. Cerebellar lesions:

It is caused by ataxia of extraocular muscles. Ask the patient to look laterally, upwards or downwards. The rapid component occurs in the direction of looking.

iii. Defective vision:

Nystagmus occurs in blind persons.

It has no rapid or slow components; but it is pendular.

It is called searching nystagmus.

2) Vertigo

- Definition: it is the sense of rotation with respect to the environment.
The person may lose balance and fall.
- Direction: it is opposite to the direction of displacement of endolymph.
 - At the start of rotation: it is in the same direction of rotation.
 - At the end of rotation: it is opposite to the direction of rotation.
- Mechanism: see mechanism of SCC stimulation.
- Pathway: see before.

3) Autonomic effects

- Definition: they are the autonomic responses to rotation as nausea, vomiting, CV changes and sweating.

- Pathway: see before.

4) Muscle tone changes

- Stimulation of SCCs → Ipsilateral increase and contralateral decrease of MT.
- Pathway: through vestibular connections with the supraspinal, extrapyramidal and spinal cord centers.

5) The past point test of Barney

- A person can very easily touch a point in front of him when:
 - His eyes are closed.
 - The point is central.
- If the person is rotated horizontally to the right on a revolving chair and rotation is suddenly stopped, he will feel rotation to the left.
- If, with his eyes closed, he is asked to touch the same central point in front of him, his hands will be deviated to the right as a compensation for the sense of rotation to the left.

CLINICAL APPLICATIONS

Meniere disease:

- It is due to supersensitivity of the vestibular system.
- On slight rotation, the patient suffers from exaggerated effects of stimulation of the SCCs in the form of vertigo, nausea, vomiting and severe autonomic changes as decreased ABP and sweating.

Motion Sickness:

- It is due to excessive and repetitive stimulation of SCCs due to motion of a vehicle (car, train, ship or plane) in non accustomed person → vertigo, nausea, vomiting and severe autonomic changes as decreased ABP and sweating.

The Utricle and Saccule:

- They are present in the vestibule.
- Their receptors are called the Macula or Otolith organ.
- Structure of the Macula: It is formed of columnar ciliated epithelium (hair cells); where the cilia project into a gelatinous material. Small calcium carbonate particles called Otoliths or Otoconia are embedded on the surface of the the gelatinous material.
- In normal erect position, the macula of the Utricle is in the horizontal plane, while that of the Saccule is vertical.
- Changing head position in space → bending of the Otoconia by their weight → stimulates or inhibits the hair cells.
- The hair cells in each macula are arranged in groups according to the direction of their stimulation. Some are stimulated on bending the head forwards, others on bending backwards and a third on bending to one side.

Functions of the Macula:

Receptors for orientation of the head in space:

- In the normal erect position → equal discharge from both sides.
- Bending the head to one side e.g. the right → bending the Otoconia → increased discharge from the right utricle and decreased discharge from the left → sense of bending the head to the right.

Receptors for linear acceleration:

- The Otoconia are displaced during linear acceleration to a direction opposite to the direction of movement due to their greater inertia.
- Example: when the body is suddenly moved forward → backward displacement of the Otoconia → bending backward of cilia → stimulation of groups of hair cells → the feeling of falling backwards → the person tries to correct himself by leaning forward.
- Receptors for labyrinthine righting reflexes according to head position.
- Receptors for labyrinthine statonic reflexes.
- Receptors for maintenance of posture during diving when both the visual and proprioceptive impulses are absent.

13. Physiology of basal ganglia

THE BASAL GANGLIA

The BG are nuclear masses that lie between the CC and the diencephalon at the base of the brain.

Structure: they include:

- The caudate nucleus (CN).
- The lentiform nucleus (lateral putamen and medial globus Pallidus).
- The functionally related nuclei; (RN, subthalamus and SN).

N.B. :

- The corpus striatum (CS) is formed of the CN and putamen.
- The GP is formed of external GP and internal GP parts.